

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		credit any three of the following four award (1) for any of following - rate decreases - reaction gets slower - rate is greatest at the start - reaction is fastest at the start award (1) for any of following - reactant particles are used up (over time) - concentration of reactants decreases (over time) - there are fewer reactant particles (over time) - there are more reactant particles at the start award (1) for any of following - there are fewer (successful) collisions per second - the frequency of (successful) collisions is lower - there is less chance of (successful) collisions - less hydrogen is produced award (1) for any of following - line levels out when reaction has finished / all the particles have reacted - particles (of metal / acid) have all been used up at 25s - reaction has finished at 25s / 35cm³	3			3		3
	(b)			(catalysts) increase the rate of a reaction / speed up a reaction / make a reaction faster (1) by lowering the activation energy / by lowering the energy needed for successful collisions (1)	2			2		2
				Question 4 total	5	4	1	10	3	8